

Srinadh Alugu

AI/ML Engineer

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Profile

AI/ML Engineer with 2.5+ years of experience in Data Science, Machine Learning, and Generative AI. Specialized in building LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, NLP pipelines, and scalable ML workflows. Strong expertise in embeddings, transformer architectures, MLOps fundamentals, and cloud-based AI deployment (Azure, AWS). Proven ability to design end-to-end AI systems that improve performance, reduce hallucinations, and enhance reliability in production environments.

Professional Experience

Data Scientist

Adarsha Automotive, India

Apr 2021 – Apr 2022

- Processed and analyzed 10,000+ multivariate sensor records, improving anomaly detection accuracy by 25% through statistical modeling and feature engineering.
- Automated Python-based preprocessing and EDA pipelines, reducing manual reporting time by 30% and improving workflow efficiency.
- Built regression and predictive models to evaluate powertrain performance trends, increasing diagnostic accuracy by 20%.
- Improved data integrity and validation checks, reducing inconsistent records by 18%.

Research Assistant – Anxiety Detection using Wearable Sensors

University of Siegen, Germany

Mar 2025 – Sep 2025

- Designed hybrid deep learning architecture (CNN + BiLSTM + GRU + Transformer Encoder) for multimodal behavioral modeling.
- Implemented self-supervised feature extraction and sensor fusion pipelines, improving cross-subject generalization by 22%..
- Built end-to-end pipelines including segmentation, normalization, model training, and statistical evaluation.
- Achieved 98.56% accuracy and 95.94% F1-score, outperforming baseline models by 15%, improving generalization across subjects through adaptive modeling strategies.

Research Assistant – LLM-Based Human Activity Recognition

University of Siegen, Germany

Sep 2025 – Present

- Designed adaptive LLM-based classification pipelines for multimodal inertial sensor data using zero-shot and few-shot learning.
- Evaluated reasoning capabilities of transformer-based LLMs on structured time-series data.
- Optimized prompt engineering strategies, improving classification consistency by 18%.
- Achieved 80.64% accuracy without model fine-tuning, demonstrating LLM reasoning feasibility in sensor-based tasks.

Projects

Knowledge-Graph-Enhanced Retrieval-Augmented Generation (RAG) System

- Designed RAG architecture integrating embeddings, vector stores, and prompt engineering.
- Reduced hallucinations by ~30% and improved answer relevance by 35%.
- Improved retrieval consistency by 40% through iterative prompt and retrieval optimization.

Transformer Fine-Tuning for Text Classification

- Fine-tuned transformer-based LLM using PyTorch on 4,000+ labeled samples.
- Achieved 95.67% test accuracy.
- Built full training, validation, and error analysis pipeline.

AI-Based Meta-Scheduling for Real-Time Systems

- Developed adaptive scheduling models using ANN, encoder-decoder networks, and Random Forest.
- Reduced re-scheduling latency by ~30% in time-triggered systems.
- Designed evaluation frameworks for robustness and scalability.

Education

Master of Science in Mechatronics (Specialization in AI)

University of Siegen, Germany Expected March 2026

Bachelor of Technology in Mechanical Engineering

Kakatiya Institute of Technology & Science, Hyderabad, India

Technical Skills

Programming: Python, C/C++

Deep Learning: Transformers, CNNs, RNNs, LSTM, GRU, Encoder-Decoder Architectures

Learning Paradigms: Self-Supervised Learning, Semi-Supervised Learning, Human-in-the-Loop Learning

AI Domains: Behavioral Modeling, Sensor Data Processing, Multimodal Learning, Time-Series Analysis

Frameworks: PyTorch, TensorFlow, Scikit-learn

Research Tools: Experiment Tracking, Statistical Evaluation, Reproducible Training Pipelines

Systems: Linux, Docker, Azure ML fundamentals

Certifications

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| • Generative AI Engineering with LLMs – IBM & Coursera | May 2024 |
| • Deep Neural Networks with PyTorch – IBM | Feb 2023 |
| • Building Deep Learning Models with TensorFlow – IBM | May 2023 |
| • Machine Learning – Andrew Ng (Stanford Online) | May 2022 |
| • AWS Cloud Technical Essentials – AWS | March 2024 |